

Epiphyte: a relational database framework for naturalistic neuroscience experiments

Alana Darcher¹, Rachel Rapp^{2,3}, Tamara T. Mueller⁴, Franziska Gerken⁵, Janne K. Lappalainen^{2,3}, Gert Dehnen¹, Marcel S. Kehl¹, Stefanie Liebe^{2,7}, Laura Leal-Taixé⁵, Florian Mormann¹, and Jakob H. Macke^{2,3,6}

¹ Department of Epileptology, University Hospital Bonn, Bonn, Germany ² Machine Learning in Science, Tübingen University, Germany ³ Tübingen AI Center, University of Tübingen, Germany ⁴ Institute for Artificial Intelligence in Medicine and Healthcare, Technical University of Munich, Germany ⁵ Dynamic Vision and Learning Group, Technical University of Munich, Munich, Germany ⁶ Max Planck Institute for Intelligent Systems, Tübingen, Germany ⁷ Department of Epileptology and Neurology, University Hospital Tübingen, Tübingen, Germany

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Summary

Neuroscience experiments generate increasingly large and complex datasets (Sejnowski et al., 2014). In particular, the use of naturalistic stimuli — such as movies (Aliko et al., 2020; Gerken et al., 2025; Keles et al., 2024), podcasts (Zada et al., 2025), and video games (Li et al., 2025) — has expanded as researchers aim to study brain activity under ecologically valid, dynamic conditions that better approximate real-world perception and cognition. These advances in experimental design and recording technologies introduce new challenges for data organization and analysis. Large-scale datasets often include multiple kinds of neural signals, complicating storage and retrieval. Continuous, naturalistic stimuli lack clearly defined event structures and require additional processing for alignment with neural activity, such as detailed annotations of stimulus features. Given these challenges, experiments comprised of high-dimensional neural data collected under naturalistic conditions can be cumbersome to analyze using standard, file system-based frameworks. To enable flexible analysis of such datasets, we developed epiphyte, a Python-based toolkit for database-backed workflows.

epiphyte implements a domain-specific relational database designed for analyzing spiking and field potential data recorded during continuous, dynamic stimuli. In addition to structuring and storing neural data, epiphyte provides a framework for organizing stimulus features and meta-data that simplifies the analysis of neural signals recorded during a movie, audio track, or similarly naturalistic stimulus. Our package offers Python modules for structuring neural data, managing stimulus metadata, and organizing results in a structured, queryable form. The package also includes modules for generating mock data, [documentation](#), and [tutorials](#) that guide users through setting up a database locally or remotely and building complete analysis workflows.

epiphyte includes components and tutorials that demonstrate how to:

- host a database on a virtual machine and enable remote access for collaborators,
- attach an external storage backend for large-scale data such as local field potentials,
- structure annotations of continuous and naturalistic stimuli,
- add new neural or stimulus data,
- analyze neural responses in relation to stimulus features, and
- train computational models directly from the database.

43 Dependencies include:

- 44 ▪ DataJoint ([Yatsenko et al., 2015](#))
- 45 ▪ MinIO (optional) ([Inc, 2015](#))
- 46 ▪ numpy ([Harris et al., 2020](#))
- 47 ▪ pandas ([McKinney & others, 2011](#))
- 48 ▪ matplotlib ([Hunter, 2007](#))
- 49 ▪ seaborn ([Waskom, 2021](#))

50 Statement of need

51 With the development of DataJoint ([Yatsenko et al., 2015](#)), relational databases have become
52 more accessible to neuroscientists. However, designing, deploying, and maintaining a database
53 ecosystem for experimental data remains technically challenging and time-consuming. While
54 many research projects already rely on relational backends (see [here](#)), few provide domain-
55 specific, worked examples that can be readily adapted to new datasets.

56 epiphyte addresses this gap by offering a standalone, fully functional framework tailored for
57 analysis of neural activity recorded during a concurrent naturalistic stimulus. It can be used as
58 an experimental analysis framework, adapted to specific experimental designs, or studied as a
59 reference implementation.

60 State of the field

61 Naturalistic neuroscience is expanding as a field. However, codebases and tools for analyzing
62 experiments utilizing naturalistic and continuous stimuli generally take the form of code released
63 alongside a publication ([Keles et al., 2024](#); [Wang et al., 2023](#)). epiphyte was built as a
64 standalone framework for structuring complex stimuli, namely movies, to support the analysis
65 of large neural datasets by providing experimenters with an end-to-end worked solution and
66 tools for relating the content of a complex, continuous stimulus with concurrently recorded
67 spiking and field potential activity.

68 Software design

69 epiphyte contains the following modules:

- 70 ▪ database: specification and implementation of the tailored database,
- 71 ▪ preprocessing: utilities for processing neural and stimulus data,
- 72 ▪ data: generators for creating mock neural and stimulus data.

73 Beyond the common Python data science dependencies, epiphyte is built using DataJoint, a
74 package that enables implementing and querying a MySQL database, and optionally, MinIO,
75 which stores large data objects (such as field potential recordings) in a queryable format.

76 The main design contribution is the structuring and organization of stimulus, neural, and
77 experiment-related meta-data. Movies and other naturalistic and continuous stimulus can
78 be difficult to align with concurrently recorded neural data. To allow for flexible use of
79 stimulus annotations, all stimulus-related items are structured as onset and offset times and
80 the associated content values, and all subsequent analysis interactions are built around this
81 structure, such as binning by movie frame (`epiphyte.data_preprocessing.binning`). Since
82 experiment set-ups and their constituent data formats can vary between labs, we provide
83 methods for generating a full mock dataset so users can quickly implement and use the
84 framework without adjusting to their specific data formats.

85 To increase the utility of epiphyte, we provide a set of tutorials for implementing a database
86 and centering subsequent data analysis around database interactions. We include walkthroughs

87 detailing how to:

- 88 1. launch the database locally or remotely,
- 89 2. attach a storage back-end to an existing database,
- 90 3. install Epiphyte locally or remotely,
- 91 4. configure and connect to a database,
- 92 5. design and implement a database,
- 93 6. develop an analysis workflow,
- 94 7. enable access for other users, and
- 95 8. run regular backups.

96 Research impact statement

97 epiphyte has been used as the main analysis framework for several projects, including the
98 decoding of movie content from human single neurons (Gerken et al., 2025), the organization
99 of a decade's collection of screening experiments (Darcher et al., 2024), and the analysis of
100 semantic receptive fields in concept cells (Karkowski et al., 2025).

101 AI usage disclosure

102 AI tools were used to generate type hints and to expand and unify documentation. All
103 generated type hints and documentation elements were verified via manual review.

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